

Supplementary Document: Mechanical Bolting & Bottom-Side Mounting

System Component: L-M3D "Abyss" Peripheral Connection

Specification: Maximal-Result Protocol / Bottom-Side Bolting Standard

1. Constructive Objective & Topology

To ensure a seamless, large-area contact zone on the underside of the multichip module carrier (PCB), the mechanical fixation of the cooling system must not cause any installation space conflicts above the motherboard. The cooling system is therefore bolted completely independently through the bottom side of the motherboard (PCB rear side). This ensures a homogeneous contact pressure distribution on the sealing surfaces (ConFlat copper seals) and prevents local shear stresses on the sensitive silicon die and the Si₃N₄ ceramic substrates.

Mechanical Layer Model of the Bolting (From Motherboard Bottom Upwards):

1. Screw head with Belleville spring washers (Stress relaxation compensation)
2. Isolation and pressure distribution Backing-Plate (Steel 1.4301 / PEEK insulator)
3. Motherboard PCB (Clearance holes without threads)
4. Housing base (Si₃N₄ Ceramic / AMB structure)
5. AlMg3 Aluminum cooling housing (Threaded blind holes M3)

2. Mathematical-Physical Derivation of Bolting Forces

To achieve absolute tightness of the metallic ConFlat sealing lip (Knife-Edge), the copper gasket must be plastically deformed. At the same time, the resulting tensile force of the screws must not exceed the compressive strength of the ceramic substrate and must not bend the PCB.

2.1 Required Total Preload Force (F_{V,ges})

The plastic deformation of the copper sealing lip requires a minimum contact pressure of $q_w = 120 \text{ MPa}$ on the effective sealing surface. The sealing surface is calculated from the mean circumference of the knife ring and the penetration width ($b_d = 0.2 \text{ mm}$).

With a seal diameter of $D = 30 \text{ mm}$, the contact area A_d is:

$$A_d = \pi \cdot D \cdot b_d = \pi \cdot 30 \text{ mm} \cdot 0.2 \text{ mm} = 18.85 \text{ mm}^2 = 1.885 \cdot 10^{-5} \text{ m}^2$$

The required minimum force F_d for plastic deformation is:

$$F_d = q_w \cdot A_d = 120 \cdot 10^6 \text{ N/m}^2 \cdot 1.885 \cdot 10^{-5} \text{ m}^2 = 2262 \text{ N}$$

Taking into account the hydraulic operating pressure of the jet impingement plenum ($p_{oil} = 0.8 \text{ bar} = 0.08 \text{ MPa}$), a hydrostatic lift-off force F_{hydro} acts on the inner chamber area ($A_{inner} \approx 500 \text{ mm}^2$):

$$F_{hydro} = p_{oil} \cdot A_{inner} = 0.08 \text{ N/mm}^2 \cdot 500 \text{ mm}^2 = 40 \text{ N}$$

With a safety factor of $S = 1.5$ against fatigue and pressure surges, the necessary total preload force $F_{V,ges}$ is calculated as:

$$F_{V,ges} = S \cdot (F_d + F_{hydro}) = 1.5 \cdot (2262 \text{ N} + 40 \text{ N}) = 3453 \text{ N}$$

With a symmetrical arrangement of 4 bolting points, each screw accounts for an axial minimum preload force of:

$$F_V = \frac{F_{V,ges}}{4} = \frac{3453 \text{ N}}{4} = 863.25 \text{ N} \rightarrow \text{Chosen: } F_V = 900 \text{ N}$$

2.2 Torque Calculation (M_A) according to Kellermann-Klein

The required tightening torque M_A is determined for the selected M3 screws (strength class 8.8). The derivation takes into account the thread pitch ($P = 0.5 \text{ mm}$), the pitch diameter ($d_2 = 2.675 \text{ mm}$), the coefficient of friction in the thread ($\mu_g = 0.12$) and at the head bearing surface ($\mu_k = 0.10$ for PEEK washer), as well as the mean head friction diameter ($D_{km} = 5.0 \text{ mm}$).

$$M_A = F_V \cdot \left[0.159 \cdot P + 0.577 \cdot d_2 \cdot \mu_g + \frac{D_{km}}{2} \cdot \mu_k \right]$$

Inserting the values:

$$M_A = 900 \cdot \left[0.159 \cdot 0.5 + 0.577 \cdot 2.675 \cdot 0.12 + \frac{5.0}{2} \cdot 0.10 \right]$$

$$M_A = 900 \cdot [0.0795 + 0.1852 + 0.2500]$$

$$M_A = 900 \cdot 0.5147 \text{ mm} = 463.23 \text{ Nmm} = 0.463 \text{ Nm}$$

Result: A precise tightening torque of **0.46 Nm** per screw guarantees plastic seal deformation without risk of damage to the threads in the AlMg3 cooling block.

3. Protection against Material Fatigue: Belleville Washers

To compensate for the thermal settlement of the components and the O-ring or copper seal relaxation during alternating load cycles, a Belleville spring washer combination is installed under each screw head.

The spring characteristic ensures that even with a settlement amount of $\Delta s = 0.05 \text{ mm}$, the preload force does not fall below the critical limit. The remaining force after heating is secured via the spring constant $C_f = 4000 \text{ N/mm}$:

$$F_{V,rest} = F_V - C_f \cdot \Delta s = 900 \text{ N} - (4000 \text{ N/mm} \cdot 0.05 \text{ mm}) = 900 \text{ N} - 200 \text{ N} = 700 \text{ N}$$

Since $4 \cdot 700 \text{ N} = 2800 \text{ N}$ is still well above the hydrostatic and plastic minimum limit of 2302 N, the system remains absolutely leak-free over its entire service life.

4. Technical Specification & Bill of Materials (BOM)

Component	Specification	Material	Quantity
Cylinder Head Screw	M3 x 16, DIN 912, ISO 4762	Steel 8.8, zinc-plated	4 pcs.
Belleville Spring Washer	3.2 x 6.0 x 0.4 mm, DIN 2093	Spring Steel 1.4310	4 pcs.
Backing-Plate	24 x 24 x 2.0 mm, CNC-milled	Stainless Steel 1.4301	1 pcs.
Isolation Washer	M3 flat washer for galvanic isolation	High-temp PEEK Polymer	4 pcs.

5. Quality Assurance and Assembly Process

- Cleaning:** Remove cutting oil residues from all threaded holes in the AlMg3 block using isopropanol.
- Pre-Assembly:** Insert the OFHC copper ring. Place the cooling block on the top of the housing.
- Backing-Plate:** Position the PEEK washers and the stainless steel backing plate on the underside of the PCB.
- Tightening Procedure:** The screws are tightened crosswise in two stages (Stage 1: 0.2 Nm; Stage 2: 0.46 Nm) using a calibrated torque screwdriver.